The general practice today is either to weld the hub liners to the face of the driving box or pour the brass or bronze on the face which has been specially prepared by machining circular dovetail grooves to hold the metal. Fig. 119 shows a specially designed turntable intended to facilitate the welding of hub liners to driving boxes. Here the large turntable may be rotated to bring each box under the hood and each box in turn may be rotated on the small individual turntable so that the welder may work around the liner without changing his position.

The shoe and wedge faces of the driving box may be machined on the planer, draw-cut shaper or milling machine. The choice of machine depends to a certain extent upon the quantity of driving box work to be done.

Fig. 120 shows a milling machine designed for production work on driving boxes. This machine has two heads, which makes possible the simultaneous machining of the shoe and wedge face.

The shoe and wedge faces should be machined so that they are at equal distances from the brass fit in the driving box (See Fig. 131). When machining shoe and wedge faces it is important, that the cutting tool closely follow the original contour of the box flanges, so that the "rock" of the box is accurately maintained. Fig. 131 in the side view illustrates what is meant by "rock" in a driving box. It will be noted that the flanges are machine straight for a specified distance in the center and that they taper outward from this straight section toward the outer ends.

The word "rock" in this instance alludes to the rocking motion of the driving box permitted by the taper on the flanges. When passing over uneven track or when for any reason the wheel on one side of the locomotive is higher than the wheel of the same set on the opposite side, the boxes, because of the manner in which the flanges are machined, may rock slightly, thus preventing broken flanges on the boxes or on the shoes and wedges.

After the brasses are pressed in, the brass retainer plugs driven in place, the shoe and wedge faces squared and the hub liners applied the boxes are ready to be bored for the journal fit. Almost all shops use either the vertical turret lathe or the special driving box boring mill for boring the crown brass and facing the hub liner. On either of these machines the two operations may be performed simultaneously. Fig. 131 shows the boring of the crown brass and facing of the hub liners on a vertical turret lathe. Reference to Fig. 131 will be helpful in following machining operations on the crown brass.

It will be noted that the crown brass is bored to the exact size of the axle journal and that the bore is slightly more than one-half the diameter of the journal. It is obvious therefore that without some further work the brass would not fit over the journal. Hence, it is necessary that the brass be relieved slightly at the outer edges. This may be accomplished by moving the box back on the boring mill table a slight distance, enlarging the bore slightly and taking a light cut down these edges. This cut should extend back just far enough to enable the brass to slip on the journal. Fig. 122 shows this relief cut only on the back edge of brass. These edges may be relieved on the shaper or in case of necessity may be chipped and filed. The hub end of the brass should be provided with a rather generous radius. (See Fig. 124.)

Wheel Centers. The wheel centers used on modern locomotives are made of cast steel but vary considerably in design. Fig. 85 shows a light weight spoke type wheel center of high tensile steel. Figs.86 and 87 illustrate the Boxpok design of wheel center. As the name implies, this center has box-section spokes and rim, the details of which are shown in Fig. 87. In Fig. 88 the disc type driving wheel center is illustrated, while the numbered details of the conventional spoke type center are given in Fig. 89.

The machining of new wheel centers may be performed largely on the heavy duty boring mill. The casting is first leveled on the machine table with the inside face up. In this position the outside diameter may be rough turned and the axle fit rough bored. The hub and upper edge of the rim may be roughed and finish cut at this time, the hub being counterbored for the wearing plate or hub liner. The casting is then turned over on the machine table and the outside hub face and rim edge roughed and finished. In this position the outside diameter and axle fit are finish turned and bored. The last operation is facing the counterbalance, which in some cases requires that the casting be set off center so that the tool will clear the crank pin boss. In some shops the counterbalance surface is machined on the planer.

The keyways are machined on the slotter and the crank pin hole is bored on the quartering machine after the wheel center is pressed on the axle.

Old wheel centers require machining under certain circumstances; if a new axle is to be applied the axle seat must be rebored on the boring mill. In this case it is considered good practice to re-turn the outside diameter in the wheel lathe after the centers have been pressed on the axle. In most instances the crank pin is removed when a new axle is applied, in which case the crank pin hole is re-bored in the quartering machine. When the new crank pin is applied it should also be finish turned in the quartering machine.

Wheel centers should be checked for diameter, taper, rotundity and concentricity, and any irregularities should be corrected before tires are applied.

Old wheel centers sometimes become out of round or develop flat spots, and should be trued up when 1/16" out of round or tapered. When truing up rims, wheel centers should not be reduced more than 1/2" below the original diameter.

Some roads use the welding process to build up low spots in wheel centers. Other roads true up wheel centers in the wheel lathe, thus reducing the diameter until certain specified limits are reached, at which time the wheel centers are restored to their original diameters by building up with fusion welding or by the application of steel bands welded to the peripheries of the rims. In some instances wheel lathes are equipped with automatic electric welding attachments for building up the rim faces of flattened or undersized wheel centers.

Fig. 90 illustrates the A.A.R. recommended practice for banding cast steel wheel centers which by successive turnings have been reduced to limit diameters. The following proceedure is advised:

True up center and cut off sufficient metal so that liner will be at least 3/8" thick after it is applied and turned.

Apply 1/2" steel liners in three sections as shown at A, Fig. 90.

When checking the tires it is essential to consider all of the tires of the set because when the tires are turned (machined in the wheel lathe) it is important that all wheels in the set have approximately the same diameter. When all tires are turned or new tires applied, the Interstate Commerce Commission rules require that there be not more than 3/32" variation in diameter of any of the wheels in the same driving wheel base. Therefore, if it is found that one tire is considerably smaller than the others it may be advisable to replace that tire with one which more nearly matches the others. This will eliminate the necessity of removing an excessive amount of service metal from the other tires in the set.

The journals should be checked to determine whether they may be returned without reducing their diameters below the shop limit. If the axle is to be renewed this should be done before proceeding with the other operations. The same proceedure should be followed with the crank pins.

If the size of the axle is suitable to continue in service, the wheel set should be sent to the journal lathe. Here the journals are turned and rolled or ground. Fig. 81 shows the journal bearings being turned in a journal turning lathe.

After the journals have been turned the wheel set may be sent to the wheel lathe or quartering machine.

Successful operation of the locomotive depends to a large extent upon accurate parallelism between the axes of the crank pins and the driving axles, accurate quartering, accurate positioning for stroke, and the absence of taper on the crank pins.

Because of these factors a quartering machine is considered essential even in the smaller shops. Fig. 82 illustrates a quartering machine set up for boring the crank pin hole in a wheel center and Fig. 83 shows detail of headstock when turning crank pin. Machining the crank pins and crank pin holes on the quartering machine eliminates all possibility of error, insures all surfaces being parallel and the driving wheels being in perfect quarter and stroke.

When it is necessary to renew a crank pin, the old pin is pressed out, using the hydraulic press. The pin hole is then rebored on the quartering machine, the new pin is pressed in and the wheel set is returned to the quartering machine where the pin is finish turned and rolled.

In some shops it is the practice, after finishing journals and crank pins, to send the sizes to the driving box and rod departments. In other shops the boring mill operator measures each journal or crank pin and bores the driving boxes or rod brasses accordingly.

The wheel sets are now sent to the wheel lathe where the tires are turned. The smallest set of tires are turned first and the size established, the remaining sets of tires for the particular locomotive are then turned to this size. The contour of the tires is turned to standard gages which show correct taper, height, thickness of flange, etc. Fig. 84 shows tires being turned on the wheel lathe.

Assuming that the work on the journals, crank pins, and tires is finished, the wheel sets are ready to have the driving boxes assembled on the axles, after which the complete units may be returned to the erecting shop.